

Dataset Construction

Construct a balanced dataset by combining **these** datasets:

- **MFakeBench** → for the **manipulated / fake classes**.
- **NewsCLIPpings** → for **real samples**.
- **DGM4** → for class 4 (if you need more samples and you can oversample MFakeBench)

Create a balanced dataset where real and fake samples are equal, and each fake scenario contains the same number of samples. The five scenarios are: **(1) Real text with real image (out-of-context), (2) Fake text with real image, (3) Real text with fake image (Photoshopped image or AI-generated image), (4) Real text with real image (genuine), (5) fake text with fake image (Photoshopped image or AI-generated image).**

If the smallest scenario (fake) has 1,000 samples, collect exactly 1,000 samples from each of the other four scenarios.

Pipeline Versions

V1: Semantic Baseline: FND-CLIP (code available)

Reference paper: "Multimodal Fake News Detection via CLIP-Guided Learning" — Zhou et al., IEEE ICME 2023.

2.1 Visual Feature Extraction

- **A pretrained ResNet-50** extracts deep spatial features from the image, producing a visual feature vector of dimension 2048. ResNet-50 is pretrained on ImageNet and fine-tuned during training.

2.2 Textual Feature Extraction

- **A pretrained BERT-base-uncased model** encodes the caption. The [CLS] token representation (dimension 768) serves as the sentence-level semantic embedding.

2.3 CLIP Cross-Modal Alignment

- **Two paired CLIP encoders** (one for image, one for text) independently embed the image and caption into a shared semantic space (dimension 512 each).

- **A cross-modal similarity score** is computed as the cosine similarity between the CLIP image and text embeddings. This score quantifies how well the image and text semantically align.
- The CLIP-generated features are concatenated into a fused multimodal vector. The similarity score is used to reweight this fused vector — when the image and text are poorly aligned, the fused features contribute less to the final prediction. This is the key innovation of FND-CLIP.

2.4 Modality-wise Attention and Classification

- A learned attention module produces three scalar scores — one for the text stream, one for the image stream, and one for the CLIP-fused stream — and computes a weighted combination of all three.
- This adaptive weighting allows the model to rely more on text for text-heavy manipulation and more on image features for visual manipulation, depending on the specific sample.
- **The final representation is passed to a two-layer MLP classifier** with a sigmoid output for binary classification (real / fake).